Information about the exam

Welcome to the exam for Secure Computer Systems I! This year's exam is a multiple choice exam with 22 graded questions, graded as pass or fail. **The maximum score is 35 points. To pass, you need at least 22 points.**

The 22 exam questions are divided into sections, roughly corresponding to the lecture topics. For each question where you can choose multiple alternatives, to discourage from simply choosing all alternatives, **incorrect alternatives will be penalized** with -1 points. However, to encourage you to answer every question, negative points don't carry over to other questions: **each question gives in the worst case 0 marks**.

Example:
If there are 2 correct answers (one point each) and 3 incorrect answers (minus one point each) and you choose all 5 answers, the question will result in points marks (not minus one!).

In other words, the exam is designed to encourage answering, but to discourage from simply guessing.

We strive to grade the exam fairly—at the end of the exam, you can write comments in a dedicated (ungraded) question to let us know if there is a problem/ambiguity in the exam questions, and justify why your answers are correct.

Good luck!

1 Confidentiality

The goal of the property *confidentiality* is:

Select all correct alternatives:

- [ ] To protect something from being modified
- [ ] To make sure something can be accessed
- [ ] To verify someone's identity
- [ ] To keep something secret
- [ ] To add redundancy to something
- [ ] To encrypt something
- [ ] To hash something

Maximum marks: 1

2 Integrity

The goal of the property *integrity* is:

Select all correct alternatives:

- [ ] To hash something
- [ ] To protect something from being modified
- [ ] To encrypt something
- [ ] To keep something secret
- [ ] To verify someone's identity
- [ ] To add redundancy to something
- [ ] To make sure something can be accessed

Maximum marks: 1

3 Availability

The goal of the property *availability* is:

Select all correct alternatives:

- [ ] To verify someone's identity
- [ ] To keep something secret
- [ ] To hash something
- [ ] To encrypt something
- [ ] To add redundancy to something
- [ ] To protect something from being modified
- [ ] To make sure something can be accessed

Maximum marks: 1

4 Design principles

Saltzer and Schroeder's design principles from 1975 are still relevant guidelines for designing secure systems. The principles are:

- Economy of mechanism
- Fail-safe defaults
- Complete mediation
- Open design
- Separation of privilege
- Least privilege
- Least common mechanism
- Psychological acceptability

Which of the following statements about the design principles are true?

Select all correct alternatives:

- [ ] Open design: widely accepted design patterns should be employed when implementing security mechanisms.
- [ ] Least common mechanism: all users should share the same mechanisms and resources.
- [ ] Separation of privilege: multiple independent conditions or privileges should be required to access critical resources.
- [ ] Fail-safe defaults: default settings should be restrictive.
- [ ] Psychological acceptability: security features should be as transparent as possible, allowing the user to decide if they want to use them or not.
- [ ] Complete mediation: access control decisions should be made primarily when a user enters a system.

Maximum marks: 2

5 Salting

What is the purpose of *salting* a password?

Select all correct alternatives:

- [ ] To allow the password to be stored as plaintext
- [ ] To make the password longer
- [ ] To increase the time it takes to check a password, thus protecting against brute-force attacks.
- [ ] To increase the likelihood of the password hash being unique

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Maximum marks: 1

6 Asymmetric and symmetric encryption

A group of 10 users want to establish secure channels among themselves. This means, every two users should be able to communicate in a way that is indecipherable by the other users. For this purpose, they are deciding between using either symmetric or public-key (asymmetric) cryptography. Which of the following statements are true?

Select all correct alternatives:

- [ ] A user would have to remember more secret keys if symmetric cryptography was used
- [ ] Symmetric cryptography requires fewer keys in total
- [ ] Adding a new user to the group is easier for public-key cryptography (i.e., it entails generating fewer keys)
- [ ] Leakage of a secret key would compromise more secure channels if symmetric cryptography is used
- [ ] Asymmetric encryption algorithms are generally faster

Maximum marks: 3

7 Replay attacks

What are important defense techniques or properties against a *replay attack*?

Select all correct alternatives:

- [ ] Hashing
- [ ] Challenge-response protocols
- [ ] Freshness
- [ ] Authentication
- [ ] Integrity

Maximum marks: 2

8 Signatures

Signatures uses public/private key pairs. When Alice wants to use such public/private key pairs when communicating with Bob, which statements are true?

Select all correct alternatives:

- [ ] Encrypting a message from Alice to Bob requires Bob's private key
- [ ] Encrypting a message from Alice to Bob requires Bob's public key
- [ ] Sending a signed message from Alice to Bob requires Bob's private key
- [ ] Decrypting a message from Bob requires Bob's private key
- [ ] Verifying a signature from Bob requires Bob's public key
- [ ] Sending a signed message from Alice to Bob requires Alice's private key
- [ ] Verifying a signature from Bob requires Bob's private key

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Maximum marks: 3

9 Access control models

Mandatory access control (MAC) and d*iscretionary access control* (DAC) are two different access control models. Which of the following statements are **not** true?

Select all false alternatives:

- [ ] In DAC, the owner decides who has access to their object
- [ ] In DAC, access rights are defined centrally
- [ ] In MAC, access rights are defined centrally
- [ ] In MAC, the owner decides who has access to their object

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Maximum marks: 2

10 Capability lists vs access control lists

An *access control matrix* can be implemented using c*apability lists* (CL) or a*ccess control lists* (ACL). CLs are per-subject lists of <object, set of rights>, and ACLs are per-object lists of <subject, set of rights>. **Which is better, when?**

Select all correct alternatives:

- [ ] CLs are better to quickly see who has access to each object
- [ ] ACLs often allow delegation of rights between subjects
- [ ] ACLs are better when subjects often need to have all their rights revoked
- [ ] CLs give a better overview of which subjects have access to a particular object
- [ ] ACLs are better when the owner of an object often needs to revoke access rights to it

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Maximum marks: 1

11 Injection attacks

Injection attacks are common software attacks, for example SQL injections. Which of the following are effective defenses against injection attacks?

Select all correct alternatives:

- [ ] Treating input as data
- [ ] Input sanitization on the client-side
- [ ] Treating input as code
- [ ] Input sanitization on the server-side
- [ ] Turning off error messages
- [ ] Using strong passwords

Maximum marks: 2

12 Vulnerable C code

The C program (Listing 1) executes a *sensitive* function only if the user provides the correct 4-digit number. The constant *PIN* stores the correct PIN, but its contents are unknown, other than the fact that it is a 5-byte string in which a 4-digit number is stored. The binary is compiled for a 32-bit architecture.

Which line introduces a buffer overflow vulnerability in the given program?

```
1  #define PIN "..." // 5-byte string storing 4-digit number
2  void sensitive () {
3   // . . .
4  }
5  int main() {
6   char secret[] = PIN;
7   char input[5]; // make room for end of string char '\0'
8   printf("Input 4-digit PIN: \n");
9   gets(input);
10  if (strcmp(input, secret) == 0) {
11   printf("Authenticated successfully\n");
12   sensitive();
13   return 1;
14  } else {
15   printf("PIN is incorrect\n");
16   return 0;
17   }
18 }
```

Listing 1: Vulnerable progam

Select one alternative

- Line 2
- Line 6
- Line 7
- Line 8
- Line 9
- Line 10
- Line 11
- Line 15

Maximum marks: 3

13 Malformed packets

Many network attacks include using *malformed* packets. What can cause malformed packets in IPv4?

Select all correct alternatives:

- [ ] Outdated routing protocols
- [ ] Intentional packet manipulation
- [ ] Improper firewall configurations
- [ ] Weak encryption schemes
- [ ] Network congestion

Maximum marks: 1

14 Distributed denial-of-service (DDoS) attacks

Protecting against *distributed denial-of-service* (DDoS) attacks based on address spoofing is difficult. What defensive mechanism most often provides effective protection?

Select all correct alternatives:

- [ ] Using DNSsec
- [ ] Upgrading your network interface to handle more incoming traffic
- [ ] Encrypting traffic
- [ ] Let endpoint routers filter packets with spoofed addresses when they enter or exit a local network

Maximum marks: 1

15 Malware type

What is a malware that replicates itself but typically does not alter its behavior called?

Select all correct alternatives:

- [ ] Virus
- [ ] Worm
- [ ] Ransomware
- [ ] Trojan

Maximum marks: 1

16 Zero-day

What is a *zero-day* vulnerability?

Select all correct alternatives:

- [ ] A vulnerability that is unknown to the vendor
- [ ] A vulnerability that gets exploited fast
- [ ] A vulnerability that has been patched
- [ ] A vulnerability that is known to the vendor

Maximum marks: 1

17 Sandboxing

Sandboxing is a defensive mechanism to improve security. Which of the following is the design principle (by Saltzer and Schroeder) that *best* addresses sandboxing?

Select the most fitting alternative:

- Open design
- Fail-safe defaults
- Least common mechanism
- Economy of mechanism

Maximum marks: 1

18 AppArmor design principle

AppArmor ("Application Armor") is a Linux kernel security module that allows the system administrator to restrict programs' capabilities with per-program profiles. Which of the following *design principles* (Saltzer and Schroeder) does AppArmor primarily address?

Select the most relevant design principle:

- Economy of mechanism
- Open design
- Complete mediation
- Least common mechanism

Maximum marks: 1

19 Online privacy

Suppose you want to avoid being tracked when surfing the web. Which of the following are **not** effective ways to protect you from being tracked?

Select all ineffective alternatives:

- [ ] Blocking social media buttons
- [ ] Using a browser with default settings
- [ ] Disabling JavaScript
- [ ] Browsing in incognito mode
- [ ] Using the Tor browser
- [ ] Using a popular search engine
- [ ] Running antivirus software

Maximum marks: 3

20 Anonymization

Privacy-preserving data analysis often involves *anonymization* to guarantee the confidentiality of inputs (data) when releasing some statistics. What are some common reasons for why anonymization is difficult and sometimes results in *re-identification*?

Select all correct alternatives:

- [ ] Anonymization only works on small data sets
- [ ] Anonymization only works on large data sets
- [ ] Lack of a proper threat model
- [ ] The existence of background data
- [ ] Anonymization algorithms are computationally expensive

Maximum marks: 2

21 Ethical hacking

Which of the following is *not* a principle of ethical hacking?

Select all alternatives that are not part of ethical hacking:

- [ ] Causing no harm to the system
- [ ] Abiding the law
- [ ] Disclosing the vulnerability publicly
- [ ] Obtaining permission in advance

Maximum marks: 1

22 Intentional flaws

What is the term for a security flaw that is *intentionally* left in the system?

Select all correct alternatives:

- [ ] Backdoor
- [ ] Weakness
- [ ] Exploit
- [ ] Vulnerability

Maximum marks: 1

23 Comments

Feedback/comments on the exam design.

In case you think something is incorrect or ambiguous in the exam design, please fill in your comments here and we will take it into account when grading:

Maximum marks: 0